# Transition Path Sampling for Boltz-2 Reverse Diffusion

Extended state space, kernels, and discrete-time TPS

**Abstract**

We formulate Boltz-2's reverse diffusion sampler as a discrete-time stochastic dynamical system with an explicit transition kernel on an extended state that includes SE(3) augmentation and Gaussian noise draws. We give path probabilities, Jacobian factors for acceptance ratios, a discussion of SDE versus Markov-chain viewpoints, and shooting moves including backward re-noising for non-equilibrium generative dynamics.

## 1 Setup and extended state space

Let $M$ denote the number of (padded) atoms. The Boltz-2 reverse sampler advances a coordinate tensor $x \in \mathbb{R}^{M\times 3}$ along a fixed schedule of noise levels $\sigma_0 > \sigma_1 > \cdots > \sigma_L = 0$ with $L =$ `num_sampling_steps`. The natural "time" parameter is the *diffusion step index* $i \in \{0, 1, \ldots, L\}$, not physical time.

At step $i$ (before applying the transition from level $\sigma_i$ toward $\sigma_{i+1}$ in the code's indexing of the loop), we record an extended state comprising:

- coordinates $x_i \in \mathbb{R}^{M\times 3}$;
- a rotation $R_i \in \mathrm{SO}(3)$, sampled via normalized quaternions mapped to matrices (Haar measure on SO(3), up to numerical implementation);
- a translation $\tau_i \in \mathbb{R}^{1\times 3}$ with $\tau_i \sim \mathcal{N}(0, s_{\text{trans}}^2 I)$ (see `compute_random_augmentation` in Boltz);
- additive noise $\varepsilon_i \in \mathbb{R}^{M\times 3}$;
- the step index $i$.

States A and B for TPS are defined by collective variables on $x_i$ at the endpoints (e.g. high noise at A, structural quality at B).

## 2 Transition kernel with SE(3) augmentation

Following the sampling loop in Boltz's `AtomDiffusion.sample`, each iteration performs:

1. **Centering and SE(3) augmentation.** Let $\bar{x}_i$ be the per-structure mean of $x_i$ (mask-weighted in general). Define
$$\tilde{x}_i = R_i(x_i - \bar{x}_i) + \tau_i.$$
2. **Noise injection.** With $\sigma^{\text{tm}}$ the current $\sigma$, $\gamma$ the schedule parameter, $\hat{t} = \sigma^{\text{tm}}(1+\gamma)$, and $\eta =$ `noise_scale`,
$$v_i = \eta^2\big(\hat{t}^2 - (\sigma^{\text{tm}})^2\big), \qquad \varepsilon_i \sim \mathcal{N}(0, v_i I), \qquad x_i^{\text{noisy}} = \tilde{x}_i + \varepsilon_i.$$

3. **Denoising update.** Let $f_\theta$ denote the preconditioned score network and $\hat{x}_i = c_{\text{skip}} x_i^{\text{noisy}} + c_{\text{out}} f_\theta(c_{\text{in}} x_i^{\text{noisy}}, \ldots)$. With $s = \texttt{step_scale}$ and $\sigma^{\text{t}}$ the next $\sigma$,

$$x_{i+1} = x_i^{\text{noisy}} + s\,(\sigma^{\text{t}} - \hat{t})\,\frac{x_i^{\text{noisy}} - \hat{x}_i}{\hat{t}}.$$

Conditioned on $x_i$, the randomness is $(R_i, \tau_i, \varepsilon_i)$ with

$$p(x_{i+1}, R_i, \tau_i, \varepsilon_i \mid x_i) = \mu_{\text{Haar}}(R_i)\,\mathcal{N}(\tau_i; 0, s_{\text{trans}}^2 I)\,\mathcal{N}(\varepsilon_i; 0, v_i I)\,\delta\big(x_{i+1} - T_i(x_i, R_i, \tau_i, \varepsilon_i)\big),$$

where $T_i$ is the deterministic map described above.

**Option A (full SE(3)).** Store $(R_i, \tau_i)$ at every step. This avoids assuming exact SE(3) equivariance of the network. Successive frames live in different rigid frames; post-alignment (e.g. Kabsch) is for analysis only.

**Acceptance ratios.** Haar density is constant on SO(3); translation densities are Gaussian and cancel when $(R_i, \tau_i)$ are resampled i.i.d. in shooting moves. Paths retain augmentation; ratio simplifications mirror the noise terms.

**SE(3) inversion for backward moves.** Given $(\tilde{x}_i, R_i, \tau_i)$, $x_i = R_i^\top(\tilde{x}_i - \tau_i) + \bar{x}_i$.

**Remark (Option B).** If the model were exactly equivariant, one could fix $R_i = I$, $\tau_i = 0$ without changing marginal path distributions; we do not rely on this here.

## 3 Path probability and Jacobians

Let $\rho(x_0)$ be the initial density (e.g. Gaussian at $\sigma_0$). The extended path

$$\mathcal{X} = \big((x_0, R_0, \tau_0, \varepsilon_0), \ldots, (x_{L-1}, R_{L-1}, \tau_{L-1}, \varepsilon_{L-1}), x_L\big)$$

has formal density (suppressing constants from Haar)

$$P(\mathcal{X}) = \rho(x_0) \prod_{i=0}^{L-1} \big[\mu_{\text{Haar}}(R_i)\,\mathcal{N}(\tau_i; 0, s_{\text{trans}}^2 I)\,\mathcal{N}(\varepsilon_i; 0, v_i I)\big]$$

under the deterministic constraints $x_{i+1} = T_i(\cdot)$.

**Log factors.** Terms $\log \mathcal{N}(\varepsilon_i)$ contribute $-\|\varepsilon_i\|^2/(2v_i)$ up to constants. When the map from $\varepsilon_i$ to $x_{i+1}$ is smooth and the network output is treated as fixed when differentiating w.r.t. $\varepsilon_i$ in a single step, the Jacobian is a scalar multiple of the identity on $\mathbb{R}^{3M}$, yielding a determinant factor $|1 + s(\sigma^{\text{t}} - \hat{t})/\hat{t}|^{3M}$ per step—schedule-dependent and cancelling in ratios with identical schedules.

**Recovering $\varepsilon_i$.** Given $(x_i, R_i, \tau_i, x_{i+1})$, recompute $\tilde{x}_i$, $x_i^{\text{noisy}}$ by inverting the affine dependence on $\varepsilon_i$ together with one network evaluation (see implementation).

## 4 SDE viewpoint versus discrete-time Markov chain

Boltz-2's sampler is *not* a standard Euler–Maruyama discretization of a fixed SDE: EDM preconditioning, $\gamma$-injection, and `step_scale` define a bespoke integrator, and $f_\theta$ is a black-box neural map.

Formally one may write $dx = b(x,t)\,dt + g(t)\,dW_t$ after reparameterizing "time" by $\sigma$, but this is heuristic. Discrete-time TPS is nonetheless rigorous: Mora, Walczak, and Zamponi (path sampling as Markov chains) and the extended-ensemble formulation of Falkner *et al.* apply whenever transition kernels and path weights are computable—which holds here.

## 5 TPS in the extended ensemble

Following Falkner *et al.*, for fixed path length the path ensemble weight ratio often reduces to checking reactivity $H_{AB}(X)$ when internal length weights cancel. For *non-reversible* generative dynamics, one-way forward shooting avoids backward kernels; two-way shooting requires a backward proposal $p^{\text{bwd}}$ (re-noising) and explicit path weight ratios built from Gaussian densities, as in the plan's quotient of products of forward and backward kernels times initial densities.

## 6 Shooting moves

**Forward shooting.** Fix $x_0, \ldots, x_k$, optionally perturb $x_k$, resample $\varepsilon_k, \ldots, \varepsilon_{L-1}$ (and fresh $(R, \tau)$), run the Boltz-2 map forward.

With a uniform shooting-point selector the Metropolis-Hastings acceptance for forward shooting is $\min(1,1) = 1$ for any reactive trial path: the generation probability for the new suffix equals the path weight of the new suffix, so both cancel in the ratio $\pi(X')/\pi(X) \cdot p_{\text{gen}}(X \to X')/p_{\text{gen}}(X' \to X)$.

**Backward shooting.** Fix $x_k, \ldots, x_L$; propose earlier frames by re-noising using the chosen backward kernel; include Gaussian log-density ratios in acceptance.

Let $X = (x_0, \ldots, x_L)$ be the current (accepted) path and $X' = (x'_0, \ldots, x'_k, x_k, \ldots, x_L)$ the trial path, which shares the suffix from index $k$ onwards. The Metropolis-Hastings acceptance ratio is

$$r = \frac{\pi(X')}{\pi(X)} \cdot \frac{q_{\text{bwd}}(X_{0:k} \mid x_k)}{q_{\text{bwd}}(X'_{0:k} \mid x_k)},$$

where $q_{\text{bwd}}(\cdot \mid x_k)$ is the backward (re-noising) proposal density for the prefix $0, \ldots, k$ conditioned on the fixed shooting point $x_k$. The suffix $k, \ldots, L$ cancels in $\pi(X')/\pi(X)$. Writing $\log \pi_{\text{fwd}}(X_{0:k}) \equiv \log \rho(x_0) + \sum_{i=0}^{k-1} \log \mathcal{N}(\varepsilon_i; 0, v_i I)$ (with optional Jacobian terms), we obtain:

$$\begin{aligned} \log r = &\left[\log \pi_{\text{fwd}}(X'_{0:k}) - \log \pi_{\text{fwd}}(X_{0:k})\right] \\ &+ \left[\log q_{\text{bwd}}(X_{0:k}) - \log q_{\text{bwd}}(X'_{0:k})\right]. \end{aligned} \tag{1}$$

Expanding,

$$\log r = \underbrace{(\text{lf_new} - \text{lf_old})}_{\text{path-weight prefix ratio}} + \underbrace{(\text{lb_old} - \text{lb_new})}_{\text{Hastings correction}} + (\log \rho(x'_0) - \log \rho(x_0)),$$

where lf_$*$ denotes the prefix forward-transition log density (Gaussian noise plus Jacobian) and lb_$*$ denotes the backward-kernel log density (also Gaussian in our re-noising scheme). The implementation in `backward_shooting_metropolis_bias` applies $\min(1, e^{\log r})$.

**Common sign-error pitfall.** Reversing the signs of the lf and lb terms (i.e. computing lf_old − lf_new + lb_new − lb_old) gives $\log(1/r)$ instead of $\log r$. When the correct $r < 1$ (move should be penalised), the inverted formula yields $\exp(-\log r) > 1$, which clamps to 1 and always accepts. When $r > 1$ (move should be freely accepted), the inverted formula gives $\exp(\log(1/r)) < 1$ and rejects moves that should be accepted. Both effects together completely invert the sampling, producing unphysical (exploded) terminal structures.

**SE(3) augmentation for noise recovery.** When `single_backward_step`$(x_{i+1}, i)$ runs, it samples a fresh augmentation $(R_i, \tau_i)$ and stores it on the *result* frame $x_i$. Thus, for a backward-generated frame $x_i$ (i.e. $s_0$ in the edge $(s_0, s_1)$ of the prefix), the correct augmentation to pass to `recover_forward_noise` is $s_0.R$ and $s_0.\tau$, not $s_1.R$ and $s_1.\tau$ (which belong to the next step $i+1$).

**Rescaled fixed-point iteration for backward inversion.** Backward shooting requires inverting the Boltz denoising update to recover the intermediate $x^{\text{noisy}}$ from the step output $x_{\text{out}}$:

$$x_{\text{out}} = (1+\alpha)\, x^{\text{noisy}} - \alpha\, \hat{x}, \qquad \alpha = s\,(\sigma^{\text{t}} - \hat{t})/\hat{t}.$$

The naïve fixed-point iteration $x \leftarrow (x_{\text{out}} + \alpha\, D(x, \hat{t}))/(1+\alpha)$ diverges at low noise because the preconditioned denoiser's skip connection $c_{\text{skip}} = \sigma_{\text{data}}^2/(\hat{t}^2 + \sigma_{\text{data}}^2)$ makes $D(x) \approx c_{\text{skip}}\, x$, giving a Jacobian with spectral radius $|\alpha\, c_{\text{skip}}/(1+\alpha)| > 1$ when $s > 1$. Factoring out $c_{\text{skip}}$ yields the rescaled equation

$$\gamma_{\text{eff}}\, x = x_{\text{out}} + \alpha\big(D(x) - c_{\text{skip}}\, x\big), \qquad \gamma_{\text{eff}} = 1 + \alpha\,(1 - c_{\text{skip}}),$$

with iteration $x \leftarrow \big(x_{\text{out}} + \alpha(\hat{x} - c_{\text{skip}}\, x)\big)/\gamma_{\text{eff}}$. The Jacobian is now $\alpha\,(D'(x) - c_{\text{skip}})/\gamma_{\text{eff}} \approx \alpha\, c_{\text{out}}\, c_{\text{in}}\, F'(\cdot)/\gamma_{\text{eff}}$, which is bounded and small at all noise levels. When `alignment_reverse_diff` is enabled, the rigid alignment from the forward sampling loop is applied inside the iteration to ensure the recovered $x^{\text{noisy}}$ is consistent with the forward kernel.

**Perturbation.** Small Gaussian noise at the shooting point improves mixing.

# 7 Backward proposal kernel: explicit derivation

The Boltz-2 denoising map $T_i$ is strictly one-directional ($\sigma$ decreases at every step), so its time-reversal under the same kernel is ill-defined. We construct instead an explicit *proposal kernel* $q_{\text{bwd}}(x_i \mid x_{i+1})$ that goes from a less-noisy frame back to a more-noisy one, and derive the exact Metropolis–Hastings correction required to maintain the correct stationary distribution over reactive paths.

## 7.1 Step 1: Invert the denoising map

The forward update (Section 1) satisfies

$$x_{i+1} = (1+\alpha)\, x_i^{\text{noisy}} - \alpha\, \hat{x}_i, \qquad \alpha = s\,\frac{\sigma^{\text{t}} - \hat{t}}{\hat{t}},$$

where $\hat{x}_i = c_{\text{skip}}\, x_i^{\text{noisy}} + c_{\text{out}}\, f_\theta(c_{\text{in}}\, x_i^{\text{noisy}}, \ldots)$ is the preconditioned denoiser. Given $x_{i+1}$, recovering $x_i^{\text{noisy}}$ requires solving this implicit equation. The rescaled fixed-point iteration

$$x \;\leftarrow\; \frac{x_{i+1} + \alpha\big(\hat{x}(x, \hat{t}) - c_{\text{skip}}\, x\big)}{\gamma_{\text{eff}}}, \qquad \gamma_{\text{eff}} = 1 + \alpha(1 - c_{\text{skip}}),$$

is contractive at all noise levels because the residual Jacobian is $\alpha\, c_{\text{out}}\, c_{\text{in}}\, F'(\cdot)/\gamma_{\text{eff}}$, small for typical EDM schedules (see Section 6). Convergence in 4 iterations is sufficient in practice. Denote the recovered intermediate $x_i^{\text{noisy}} \approx \mathcal{I}(x_{i+1}, i)$.

### 7.2 Step 2: Propose fresh stochastic draws

Given $x_i^{\text{noisy}}$ from step 1, draw independent samples

$$\varepsilon_i' \sim \mathcal{N}(0, v_i I), \qquad R_i' \sim \mu_{\text{Haar}}, \qquad \tau_i' \sim \mathcal{N}(0, s_{\text{trans}}^2 I).$$

These fresh draws are *independent* of the draws that produced $x_{i+1}$ in the forward pass.

### 7.3 Step 3: Construct the proposed earlier frame

Using the SE(3) inversion $x_i = R_i'^{\top}(\tilde{x}_i - \tau_i') + \bar{x}_{\text{ref}}$ with $\tilde{x}_i = x_i^{\text{noisy}} - \varepsilon_i'$, we obtain the proposed coordinate $x_i$. The reference centroid $\bar{x}_{\text{ref}}$ is taken from the input (the original $\bar{x}_i$ stored during the forward pass, if available, or set to zero).

### 7.4 Backward proposal density

Since $x_i^{\text{noisy}}$ is a *deterministic* function of $x_{i+1}$ (via the fixed-point inversion), the only randomness in the backward step is $(\varepsilon_i', R_i', \tau_i')$. The backward proposal density is therefore

$$q_{\text{bwd}}(x_i \mid x_{i+1}) = \mathcal{N}(\varepsilon_i';\, 0, v_i I)\; \mu_{\text{Haar}}(R_i')\; \mathcal{N}(\tau_i';\, 0, s_{\text{trans}}^2 I)\; |\det J_{\text{bwd}}^{-1}|, \tag{2}$$

where $J_{\text{bwd}}^{-1}$ is the Jacobian of the map $(\varepsilon_i', R_i', \tau_i') \mapsto x_i$ with $x_i^{\text{noisy}}$ fixed. Differentiating step 3 with respect to $\varepsilon_i'$ at fixed $R_i', \tau_i'$ gives $\partial x_i/\partial \varepsilon_i' = -R_i'^{\top}$, so $|\det(\partial x_i/\partial \varepsilon_i')| = |\det R_i'| = 1$. The Haar and translation Jacobians are also unit-modulus. Hence $|\det J_{\text{bwd}}^{-1}| = 1$ and the backward proposal density is simply the product of the noise Gaussian densities:

$$\log q_{\text{bwd}}(x_i \mid x_{i+1}) = -\frac{\|\varepsilon_i'\|^2}{2v_i} - \frac{\|\tau_i'\|^2}{2s_{\text{trans}}^2} + \text{const}, \tag{3}$$

where the Haar term is constant and the translation Gaussian cancels in acceptance ratios (since old and new backward draws are drawn from the same prior). The *effective* log density used in the MH ratio reduces to

$$\log q_{\text{bwd}}^{\text{eff}}(x_i \mid x_{i+1}) = -\frac{\|\varepsilon_i'\|^2}{2v_i}.$$

### 7.5 Why forward and backward densities differ

The forward density of the transition $x_i \to x_{i+1}$ is

$$\log p_{\text{fwd}}(x_{i+1} \mid x_i) = -\frac{\|\varepsilon_i\|^2}{2v_i} + \log|\det J_{\text{fwd}}|,$$

where $\varepsilon_i = x_i^{\text{noisy}} - \tilde{x}_i$ is the specific noise draw that was used, and the Jacobian $|\det J_{\text{fwd}}| = |1+\alpha|^{3M}$.

The backward proposal density at the same transition (Eq. 3) uses a *fresh independent draw* $\varepsilon_i'$ from the same prior, not the "inverse" of the denoising kernel. Because $p_{\text{fwd}}(x_{i+1} \mid x_i) \neq q_{\text{bwd}}(x_i \mid x_{i+1})$ in general, the MH ratio for backward shooting is non-trivial and requires the explicit Hastings correction derived in Section 6.

### 7.6 Full Metropolis–Hastings ratio for backward shooting

For a backward shooting move at frame $k$ (suffix $k, \ldots, L$ shared), the prefix $0, \ldots, k$ is re-proposed by applying $q_{\text{bwd}}(\cdot \mid \cdot)$ sequentially from $x_k$ backwards to $x_0$. Let $X = (x_0, \ldots, x_L)$ be the current path and $X' = (x'_0, \ldots, x'_k, x_{k+1}, \ldots, x_L)$ the trial path ($x'_j = x_j$ for $j \geq k$). Writing $\pi_{\text{fwd}}(X_{0:k}) = \rho(x_0) \prod_{i=0}^{k-1} p_{\text{fwd}}(x_{i+1} \mid x_i)$ and similarly for $X'$, and $Q_{\text{bwd}}(X_{0:k}) = \prod_{i=0}^{k-1} q_{\text{bwd}}(x_i \mid x_{i+1})$, the acceptance ratio is

$$\begin{aligned} \log r = & \underbrace{\left[\log \pi_{\text{fwd}}(X'_{0:k}) - \log \pi_{\text{fwd}}(X_{0:k})\right]}_{\text{forward path-weight ratio}} \\ & + \underbrace{\left[\log Q_{\text{bwd}}(X_{0:k}) - \log Q_{\text{bwd}}(X'_{0:k})\right]}_{\text{Hastings correction}} \\ & + \underbrace{\left[\log \rho(x'_0) - \log \rho(x_0)\right]}_{\text{initial-prior ratio}} . \end{aligned} \tag{4}$$

Expanding with Eq. 3, the Hastings correction evaluates to the *difference of Gaussian log-densities* for the re-noising draws ($\varepsilon'_i$) versus those implied by the reverse trajectory:

$$\log Q_{\text{bwd}}(X_{0:k}) - \log Q_{\text{bwd}}(X'_{0:k}) = \sum_{i=0}^{k-1} \left[ -\frac{\|\varepsilon_i^{\text{old}}\|^2}{2v_i} + \frac{\|\varepsilon_i^{\text{new}}\|^2}{2v_i} \right],$$

where $\varepsilon_i^{\text{old}}$ is the backward-step noise stored on frame $x_i$ (old path) and $\varepsilon_i^{\text{new}}$ is the noise stored on $x'_i$ (new path).

## 8 Ergodicity of the full move set

Define the *reactive path ensemble* $\mathcal{R}$ as the set of all fixed-length paths $X = (x_0, \ldots, x_L)$ with $x_0 \in A$ and $x_L \in B$, weighted by $\pi(X) \propto P(\mathcal{X})\, \mathbf{1}[X \in \mathcal{R}]$.

**Forward-only chain.** Forward shooting at index $i^*$ leaves the prefix $x_0, \ldots, x_{i^*}$ unchanged. The chain is ergodic over the sub-ensemble with fixed $x_0$ but is not ergodic over $\mathcal{R}$: paths with different starting noise configurations $x_0$ are never reached.

**Backward-only chain.** Backward shooting at index $i^*$ leaves the suffix $x_{i^*}, \ldots, x_L$ unchanged. Choosing $i^* = 1$ regenerates the entire prefix except $x_0$ itself; choosing $i^* = L - 1$ keeps only $x_L$ fixed. However, the chain is not ergodic over $\mathcal{R}$ in general because $x_L$ (the terminal structure) is never resampled.

**Forward + backward chain (two-way shooting).** Using forward and backward shooting together breaks both limitations: backward moves can change $x_0$ (by choosing $i^* = 1$) while forward moves can change $x_L$ (by choosing $i^* = L - 1$).

**Theorem 1** (Ergodicity of two-way shooting)**.** *Suppose the reactive path space is connected under the move set, i.e. for any two reactive paths $X, X'$ there exists a finite sequence of forward and backward shooting moves (with any shooting indices in $\{1, \ldots, L-1\}$) of positive probability that transforms $X$ into $X'$. Then the Markov chain on $\mathcal{R}$ defined by the forward + backward shooting*

*moves with the acceptance rules in Eqs. 4 and Section 6 is ergodic and has stationary distribution $\pi$.*

*Proof sketch.* Standard Metropolis–Hastings theory [3]: each move satisfies detailed balance with respect to $\pi$ by construction of the acceptance ratio. Irreducibility follows from the assumed connectivity of path space. □

**Global reshuffle move.** For additional robustness, a *global reshuffle* move may be included: draw $x_0' \sim \rho(\cdot)$ and all $(\varepsilon_i', R_i', \tau_i')$ fresh, then forward-integrate the full path. The acceptance ratio is

$$\frac{\pi(X')}{\pi(X)} \cdot \frac{q_{\text{reshuffle}}(X \to X)}{q_{\text{reshuffle}}(X \to X')} = \frac{\rho(x_0') \prod_i \pi_i(\xi_i')}{\rho(x_0) \prod_i \pi_i(\xi_i)} \cdot \frac{\rho(x_0) \prod_i \pi_i(\xi_i)}{\rho(x_0') \prod_i \pi_i(\xi_i')} = 1,$$

so the reshuffle always accepts (when the trial path is reactive). It reaches any path in a single step, guaranteeing irreducibility on its own. In practice it is used at $\sim$10% of MC steps to complement the local shooting moves.

**Complete move scheme.** The recommended scheme uses:

- 45% forward shooting (uniform selector in $\{1, \ldots, L-1\}$, always accepts);
- 45% backward shooting (uniform selector, explicit MH correction via Eq. 4);
- 10% global reshuffle (always accepts).

This is ergodic on $\mathcal{R}$ without any assumption about connectivity.

## 9 Collective variables

Useful CVs: RMSD to a reference, radius of gyration, contact maps, predicted pLDDT from the denoised structure, attention entropy, and the diffusion step index.

## References


[1] T. Mora, A. M. Walczak, and F. Zamponi, Transition path sampling for discrete time Markov dynamics, *arXiv*:1202.0622 (2012).

[2] S. Falkner, A. Coretti, B. G. Peters, P. G. Bolhuis, and C. Dellago, Revisiting shooting point Monte Carlo methods for transition path sampling, *J. Chem. Phys.*, arXiv:2408.03054 (2024).

[3] P. G. Bolhuis, D. Chandler, C. Dellago, and P. L. Geissler, Transition path sampling: throwing ropes over rough mountain passes, in the dark, *Annu. Rev. Phys. Chem.* **53**, 291 (2002).

[4] P. Buijsman and P. G. Bolhuis, Transition path sampling for non-equilibrium dynamics without predefined reaction coordinates, *J. Chem. Phys.* **152**, 044107 (2020).

[5] A. N. Singh and D. T. Limmer, Reactive path ensembles within nonequilibrium steady states, *arXiv*:2501.19233 (2025).

[6] Z. F. Brotzakis and P. G. Bolhuis, A one-way shooting algorithm for transition path sampling of asymmetric barriers, *J. Chem. Phys.* **145**, 164112 (2016).